

DETECTING EXTERNAL IP RISK ANALYSIS

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ABSTRACT

With the new age of rising cybersecurity attacks, external IP addresses continue to be one of the most used attack vectors for launching cyber attacks such as Distributed Denial of Service (DDoS), phishing, malware spreading, and unauthorized access attempts. With the surge in dependence on cloud-based infrastructure, remote work, and inter-connected systems, it has now become essential for organizations to deploy sound, real-time defense mechanisms. This introduces a system of complete real-time threat detection and risk analysis that continuously scrutinizes incoming network traffic to determine the threat level of external IPs engaging with enterprise resources.

The system uses real-time event logging and consumes several external threat intelligence services, such as VirusTotal, AbuseIPDB, and IPInfo, to enhance IP metadata with threat reputation, geolocation information, and historical activity profiles. The open-source security framework Wazuh is utilized as a central system for threat detection and response, and is employed to enable integration with the VirusTotal API for automated queries of threat intelligence.

Internally derived metrics—such as connection attempt frequency, access to essential ports, and suspicious traffic patterns—are compared against external intelligence to determine a dynamic risk score per IP.

High-risk IPs are automatically identified based on preconfigured thresholds, sending alerts and activating defense procedures like IP blocking or alerting security teams. This smart and automated system facilitates real-time threat detection, lowers response time dramatically, and strengthens enterprise network resilience to complex external cyberattacks.

1. INTRODUCTION

With growing complexity in digital infrastructures and the extensive use of cloud services, remote working environments, and interconnected networks, organizations face more exposure than ever before to cybersecurity attacks from external origins. Among the most common and perilous attack vectors are malicious external IP addresses trying to communicate with internal systems.

These IPs may be linked with several types of threats such as

- Distributed Denial of Service (DDoS) attacks
- Phishing attacks
- Malware distribution
- Brute-force login attempts and
- Other unauthorized access types.

With more intelligent and automated threats from cybercriminals, the old-fashioned perimeter defense is no longer enough to detect and block threats in a timely and effective manner. In order to overcome this threat, adopting a proactive strategy with constant monitoring, analysis, and reacting against external IP activity is essential. To detect and analyse the threat of an external IP calls for a mix of behavioural analysis and threat intelligence. Indicative elements like geolocation inconsistency, uncharacteristic traffic patterns, attempts to access core ports, and frequency of connections are markers of malicious behaviour.

Internal telemetry is also augmented by real-time threat feeds from trusted sources such as VirusTotal, AbuseIPDB, and IPInfo, which add further background information regarding an IP's past behavior and reputation. Wazuh, an open-source security monitoring system, is a key tool in doing so by offering centralized log aggregation, real-time analysis, and automated correlation with threat intelligence APIs like VirusTotal. Wazuh allows security teams to automate IP data enrichment, assign risk scores, fire alerts, and enact preconfigured response actions like blocking malicious IPs or alerting analysts.

This multi-level and intuitive strategy greatly enhances an organization's capacity for detecting, evaluating, and responding to external threats in a timely manner, minimizing the likelihood of security breaches and providing firmer protection of vital assets.

2. LITERATURE SURVEY

- **A Survey of Machine Learning for Big Code and Naturalness (2021)** by **P. Allamanis et al.** While focused on code analysis, this work discusses behavioural modeling and anomaly detection—approaches applicable to identifying malicious IP activity.
- **Cyber Threat Intelligence: A Review of Research and Practice (2020)** by **A. H. Lashkari et al.** This paper explores various Cyber Threat Intelligence (CTI) platforms and discusses how integrating CTI into detection systems can help identify malicious IPs.
- **Detecting Malicious IP Addresses Using DNS and WHOIS Features (2021)** by **J. Bilge and T. Dumitras.** This paper proposes a machine learning-based approach for identifying malicious IPs by analyzing DNS and WHOIS data, combined with access behaviour.
- **Fraudulent IP Address Detection Using Machine Learning Techniques (2024)** by **Singh, P, Abdul Razak, A.** It Describes three ML-based methods for detecting fraudulent IP behavior with low false positives, designed for real-time deployment.
- **Graph Neural Networks and Cross-Protocol Analysis for Detecting Malicious IP Addresses (2022)** by **Huang Y, Negrete J, Wagener J, Wosotowsky A, et al.** Proposes a supervised learning model combining email, web, and DNS features using a graph-based approach to assess IP reputation across protocols.
- **Intelligent Threat Detection Using Wazuh and Machine Learning (2022)** by **M. López and J. Ortega.** This research investigates the use of the Wazuh SIEM platform for detecting threats using both signature-based rules and integrated APIs such as VirusTotal.

- **IP Reputation for Botnet Detection in Network Traffic (2019)** by S. Zeidanloo et al. This study introduces the concept of leveraging IP reputation to identify botnet-related activities.
- **LogSHIELD: A Graph-based Real-time Anomaly Detection Framework using Frequency Analysis (2024)** by Roy, K. C., Chen, Q. Proposes a graph-based anomaly detection method on host logs with frequency domain analysis.
- **siForest: Detecting Network Anomalies with Set-Structured Isolation Forest (2024)** by Christie Djidjev.
- **HyperVision: Detecting Unknown Encrypted Malicious Traffic in Real Time via Flow Interaction Graph Analysis (2023)** by Chuanpu Fu, Qi Li, Ke Xu. This work builds a graph from network flow interactions, and uses unsupervised graph learning to detect unknown malicious traffic, especially encrypted traffic.
- **Systematic review and characterisation of malicious industrial network traffic datasets (2025)** by International Journal of Information Security. It focuses on industrial / IIoT / OT networks: examines existing datasets for malicious network traffic, categorizes attack types, analyses complexity, etc.
- **High-Speed Network DDoS Attack Detection: A Survey (2023)** by Haseeb-ur-rehman et al. It surveys techniques for DDoS detection in high-speed networks. Although it's more specific to DDoS, many methods overlap with malicious IP detection. Might be relevant if some malicious IPs are involved in volumetric attacks.

3. PROPOSED SYSTEM

The system proposed here fills the above-mentioned loopholes by providing a real-time risk analysis and detection solution specially designed to assess external IP addresses trying to or having accessed the enterprise network. It combines live network feeds with external threat intelligence, thus offering a proactive and context-based protection system.

FEATURES

- **Real-time Monitoring:** Catches and logs live connection attempts to the network via network logging mechanisms.
- **Threat Intelligence Integration:** Makes use of APIs from providers such as VirusTotal and AbuseIPDB to retrieve real-time threat data for every IP.
- **Automated Risk Scoring Engine:** Determines a composite risk score based on parameters including:
 - IP reputation from threat databases
 - Frequency of access attempts
 - Protocol misuse
 - Geolocation anomalies
 - Known associations with malicious activities
- **Suspicious IP Flagging:** Automatically flags IPs based on their risk score and sends alerts to security personnel.
- **Optional Mitigation:** Can integrate with firewall systems or SIEM platforms to block or quarantine high-risk IPs automatically.
- **Dashboard and Reporting:** Presents the risk analysis in an intuitive format for monitoring, auditing, and decision-making.

3.1 Preventive measures

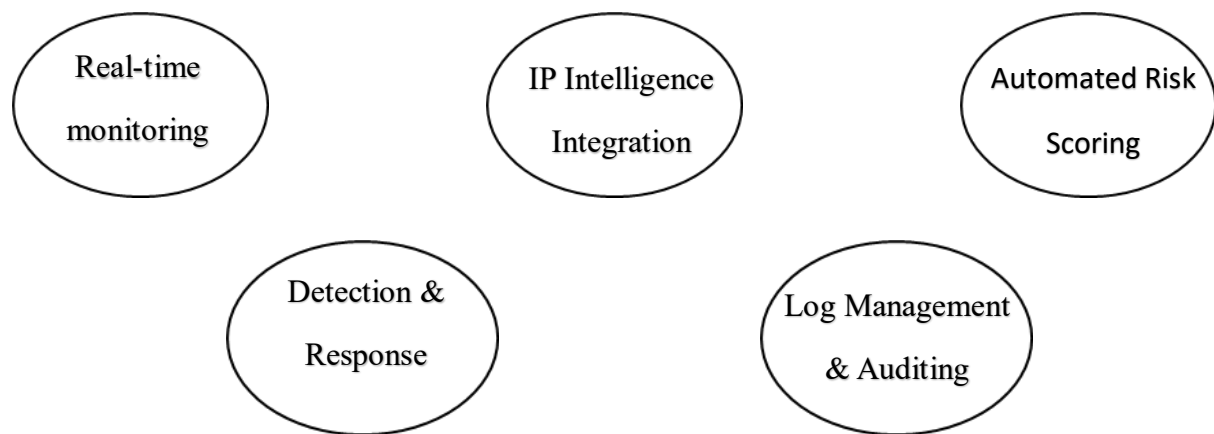


Fig.1 – Preventive Measures

1.Real-Time Monitoring

- Monitor inbound traffic in real time with network sensors.
- Examine patterns like login attempts, port scanning, and abnormal volumes.

2. IP Intelligence Integration

- Integrate external threat feeds “VirusTotal” to enhance IP profiles.
- Keep a current blacklist of IPs with known malicious activity.

3. Automated Risk Scoring

- Dynamically assign a risk score to every IP using behavioural and reputational intelligence.
- Trigger automatic defences based on this score.

4. Wazuh-Based Detection & Response

- Implement custom detection rules for abnormal behaviour.
- Integrate with the Wazuh platform for real-time alerting and firewall blocking.

5. Log Management & Auditing

- Store and analyse logs to comprehend attack patterns and facilitate compliance.
- Allow retrospective analysis of IP activity.

3.2 Key features

- **Real-Time Event Logging:** Logs and analyzes all incoming IP connections instantly for threat detection.
- **Threat Intelligence Integration:** Uses APIs from VirusTotal and AbuseIPDB, IPInfo for log collection to enrich data with known threat information.
- **Wazuh Integration:** Centralizes security monitoring and enables custom rule definitions, log analysis, and alerting.
- **Dynamic Risk Scoring:** Assigns risk scores to IPs based on frequency, behavior, geolocation, and threat history.
- **Automated Mitigation:** Blocks IPs or triggers alerts automatically when threat thresholds are exceeded.
- **Geolocation & Metadata Analysis:** Identifies the geographic origin and context of IP traffic for better decision-making.
- **Custom Rule Engine:** Allows enterprises to define what constitutes "suspicious" behaviour based on their own risk posture.
- **Dashboards & Reporting:** Provides visualizations and reports for ongoing threats, blocked IPs, and system activity.
- **Scalable and Cloud-Ready:** Works seamlessly across hybrid cloud environments with support for distributed monitoring.
- **API Integration for SIEM Ecosystems:** Supports REST API and webhook integrations to push threat data or receive instructions from Security Orchestration, Automation, and Response tools and third-party SIEMs.

3.3 Methods to raise external ip risk awareness

1. Employee Security & Education Training:

a. Periodic Cybersecurity Awareness Programs:

- Organizations should have regular training sessions in place to address threats from outside Ips.
- Also the ways attackers utilize IP addresses for malicious activity—such as reconnaissance, phishing, and DDoS.
- These should not be generic programs but role-based.

b. Interactive Training Simulations:

- In order to effectively raise awareness, organizations must incorporate hands-on, interactive exercises.
- By including simulated phishing attacks or simulated brute-force attacks from known bad IPs.
- These exercises are used to train employees to identify suspicious access attempts, identify odd traffic sources, and know what to do.

2. Technical Team Workshops & Playbooks:

a. SOC and IT Team Workshops:

- How to analyze threat intelligence obtained from platforms such as VirusTotal API, AbuseIPDB, IPInfo.
- How to interpret Wazuh alerts and logs.
- How to map IP behavior patterns like frequency of connections, geolocation anomalies onto actual threats.

b. Threat Detection Playbooks:

- Definition of a suspicious IP.
- Severity-based escalation instructions.
- Automated and manual remediation instructions blacklisting, isolation of compromised systems.

3. Internal Communication & Awareness Campaigns:

a. Security Newsletters & Threat Bulletins:

- Recent malicious IPs that have been blocked by the system.
- Trends in IP-based threats emerging.
- Success stories regarding how the system prevented breaches.

b. Awareness Posters and Digital Signage

Placing posters in shared spaces such as IT rooms or next to workstations can show reminders such as:

- "Don't trust unknown IPs!"
- "Have a security warning about an unusual IP? Report it right away."

4. Threat Intelligence Integration Awareness:

a. Educating on the Value of Threat Intelligence:

- How VirusTotal, AbuseIPDB, and IPInfo add value to IP metadata.
- Why reputation scores are important.
- How these integrations inform detection and blocking processes.
- Live Demonstrations of Detection Capabilities

b. Live Demonstrations of Detection Capabilities:

- Organize live demos where teams can see the system in action—by monitoring logs in Wazuh or showing a real-time response to a flagged malicious IP.
- For example, show how a suspicious IP triggers a VirusTotal lookup, how the system calculates a risk score, and how Wazuh pushes an alert and blocks the IP.

5. Cross-Department Collaboration & Role-Based Awareness:

a. Inter-Departmental Cybersecurity Exercises:

- Exterior IP threats tend to cross departments—marketing might become the target of phishing, while cloud infrastructure teams face DDoS attacks.
- Run cross-functional tabletop exercises or red team/blue team drills to make people aware as a group and enhance coordination.

b. Role-Based Risk Briefings:

- Executives need to know the business ramifications of exterior IP-based threats.
- Developers need to know how to protect APIs and endpoints against unauthorized IP access.

6. Metrics, Dashboards & Transparency:

a. Staff Threat Dashboards:

Deploy interactive dashboards displaying current metrics:

- Blocked IP numbers
- Most active threat geolocations
- Wazuh alerts

b. Monthly Threat Review Reports:

Share simple-to-understand monthly reports detailing:

- Top IP-based threat incidents
- Remediation steps executed

4. METHODOLOGY

The methodology focuses on designing and implementing a real-time system that detects, analyzes, and responds to external IP threats using behavioural analytics and external threat intelligence.

Steps Involved:

1. **Real-Time Log Collection:**

- Network logs are collected via Wazuh agents from endpoints, firewalls, or gateways.

2. **IP Extraction & Enrichment:**

- Extract external IPs attempting to access the network.
- Use VirusTotal API, AbuseIPDB, and IPInfo to enrich data with reputation, geolocation, and history.

3. Dynamic Risk Scoring:

- Each IP is evaluated using a scoring algorithm based on:
 - Threat intelligence score
 - Frequency of access
 - Protocol misuse
 - Port scanning behavior
 - Location anomalies

Table 1 – Dynamic Risk Scoring Algorithm

Feature	Description
Threat Intelligence	Normalized score from VT, AbuseIPDB, etc.
Frequency of Access	How often the IP attempts access in a defined time window (e.g., >50 attempts in 5 min)
Protocol Misuse	Use of uncommon or suspicious protocols (e.g., ICMP used excessively, SSH brute force attempts)
Port Scanning	Sequential port probes from the same IP; sudden increase in SYN packets without completed connections
Location Anomalies	IP from a region that doesn't match past behavior, or from embargoed/sanctioned countries (e.g., North Korea, Iran)

4. Detection and Alerting:

- Wazuh applies rules and triggers alerts when thresholds are crossed.

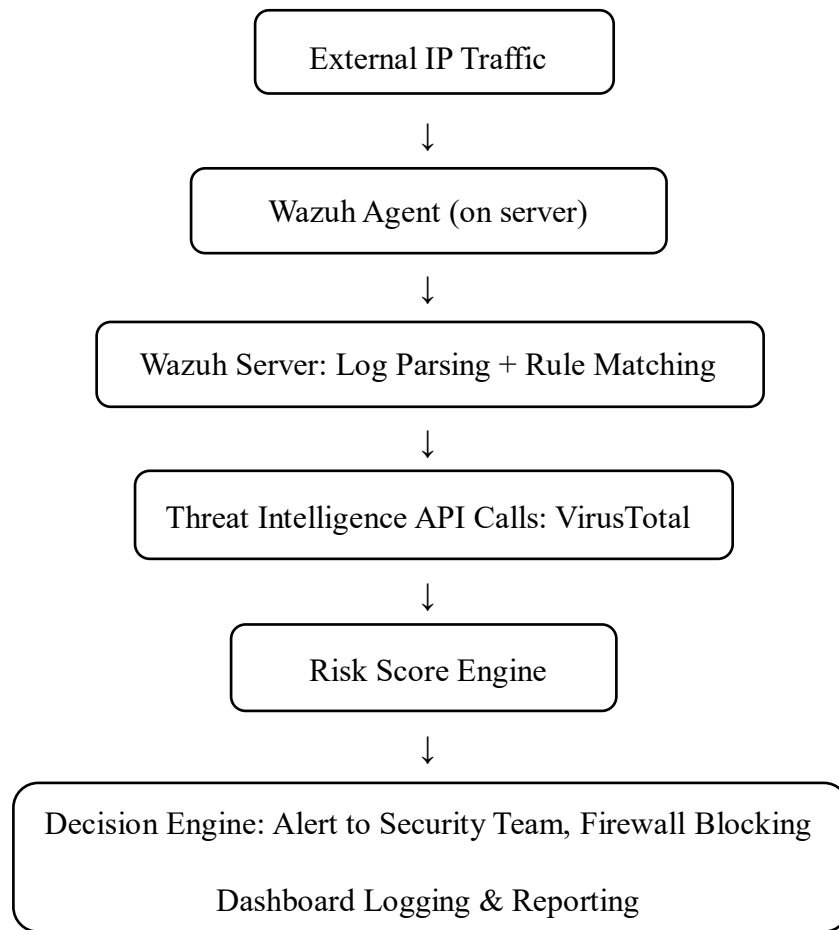
5. Automated Response:

- High-risk IPs are blocked automatically through firewall integration or SIEM actions.

6. Visualization and Auditing:

- All events are visualized through Wazuh dashboards and stored for audit purposes.

4.1 Work flow diagram



SYSTEM WORKFLOW: DETAILED EXPLANATION

1) External IP Traffic

Any incoming network traffic originating from outside your trusted internal network typically from the public internet.

Examples:

- Users accessing web servers
- API requests from third-party services
- Potential scans or brute-force attempts from unknown hosts

Security Implication:

This traffic is often the initial entry point for attacks like:

- Port scanning
- DDoS attempts and Brute-force logins
- Exploiting web app vulnerabilities

Why Monitor It?

- Attackers often probe open ports or services to find weak spots.
- Knowing who is connecting is the first step to attribution and response.

2) Wazuh Agent (on Server)

Lightweight software installed on individual hosts (servers, workstations, etc.) that continuously monitors the system for events.

Monitors:

- Network activity: source/destination IPs, ports, protocols
- System logs: authentication attempts, process creation, file modifications
- Security logs: failed logins, privilege escalation, service status changes

Use Case:

- Detects a failed SSH login attempt from an external IP
- Sends event data to the central Wazuh server for analysis

Benefits:

- Distributed monitoring across your infrastructure
- Real-time data collection ensures up-to-date visibility

3) Wazuh Server: Log Parsing + Rule Matching

Function:

- The Wazuh manager/server collects logs from all deployed agents.
- It parses the log data and compares it against predefined or custom rules to identify potential threats.

Capabilities:

- Uses decoders to understand log formats (e.g., Syslog, Windows Event Logs)
- Applies detection rules to match suspicious patterns (e.g., too many failed logins, rapid port probing)

Outcome:

- Flags the event for further analysis if a rule is matched.

Example Rule Match:

"If more than 10 failed login attempts occur from the same IP within 1 minute → raise alert."

4) Threat Intelligence API Calls – VirusTotal, AbuseIPDB, IPInfo

Enrich flagged IP addresses with contextual threat data using External threat intelligence sources.

How it Works:

Once an IP is flagged, the server automatically queries:

- VirusTotal: AV detection history, file/URL associations, community votes
- AbuseIPDB: Crowdsourced abuse reports (e.g., SSH brute force, spam, DDoS origins)
- IPInfo: ASN, geolocation, VPN/proxy usage, hosting provider

Example Outcome:

- VirusTotal reports the IP has delivered malware.
- AbuseIPDB confirms it was reported multiple times for port scanning.
- IPInfo shows it belongs to a data center in an unexpected region (e.g., North Korea).

5) Risk Score Engine

Purpose: Aggregates multiple threat signals and calculates a composite risk score for each flagged IP.

Data Sources Used:

Internal Behavior:

- Frequency of connections
- Port usage (e.g., targeting port 22 repeatedly)
- Time of access (e.g., off-hours)
- Rate of login failures

External Intelligence:

- VirusTotal detection ratio
- AbuseIPDB confidence score
- IPInfo metadata

Risk Scoring Logic:

A scoring algorithm assigns weights to each feature. For example:

$$\begin{aligned} \text{Risk Score} = & (0.3 - \text{VT Score}) + (0.2 - \text{Access Frequency}) + (0.2 - \text{Port Scan Evidence}) \\ & + (0.2 - \text{Geolocation Anomaly}) + (0.1 - \text{Historical Abuse Score}) \end{aligned}$$

Outcome:

- Low score (60): Trigger alert and possible mitigation.

Why It Matters:

- Reduces false positives.
- Avoids over-blocking legitimate users.
- Provides a quantified risk value to aid human analysts.

6) Decision Engine

Acts upon the risk score and threat context to take automated or semi-automated actions. Actions Taken:

1.Alerting:

Notifies security teams via:

- Wazuh dashboard
- Email
- Syslog forward to SIEM (e.g., Splunk, ELK)

2.Blocking:

Automatically pushes a firewall rule to block the IP on:

- Host firewall (via Wazuh Active Response)
- Perimeter firewall (e.g., pfSense, iptables)
- Cloud firewall (e.g., AWS Security Groups)

3.Logging:

- All decisions and actions are written to audit logs for future review.
- Useful for incident response and compliance audits.

4.2 How all components work together

A[External IP Traffic] --> B[Wazuh Agent Monitors Events]

B --> C[Wazuh Server Parses Logs & Matches Rules]

C --> D[Suspicious IP?]

D -- Yes --> E[Threat Intel API Calls (VT, AbuseIPDB, IPInfo)]

E --> F[Risk Score Engine Aggregates Threat Data]

F --> G[Score > Threshold?]

G -- Yes --> H[Decision Engine: Alert + Block]

H --> I[Log Action + Update Dashboards]

4.3 Benefits of this detailed process

Table 2 – Benefits of the above Workflow Diagram

Feature	Impact
Real-time detection	Respond to threats as they happen
External + internal analysis	Accurate decision-making, reduced false positives
Automated mitigation	Saves analyst time, blocks threats instantly
Customization	Fit to enterprise risk posture, custom rules, flexible scoring
Auditability	Every decision is logged and traceable for future reviews

Fig.3 – VirusTotal API Metrics

The IP address 122.179.163.160 has been identified as malicious by reputable threat intelligence providers, with conclusive evidence linking it to a range of harmful cyber activities, including the distribution of malware, involvement in suspicious or unauthorized network behavior, and indicators consistent with botnet operations suggesting that it may be part of a larger coordinated infrastructure used to compromise systems, exfiltrate data, or launch attacks across networks.



Fig.4 – VirusTotal API IP Metrics

The analysis of the suspicious IP address 122.179.163.160 uncovers a web of network relationships and global connections, revealing that it has established communication with multiple international destinations and domains. This pattern is strongly indicative of command-and-control (C2) behaviour, where the IP may be functioning as a central node directing or receiving instructions from infected systems across the globe. Such connections not only highlight the geographical spread of potential compromise but also assist in pinpointing affected regions and correlating activities with known threat actors or malware campaigns, thereby offering critical context for threat attribution and response efforts.

WAZUH DASHBOARD

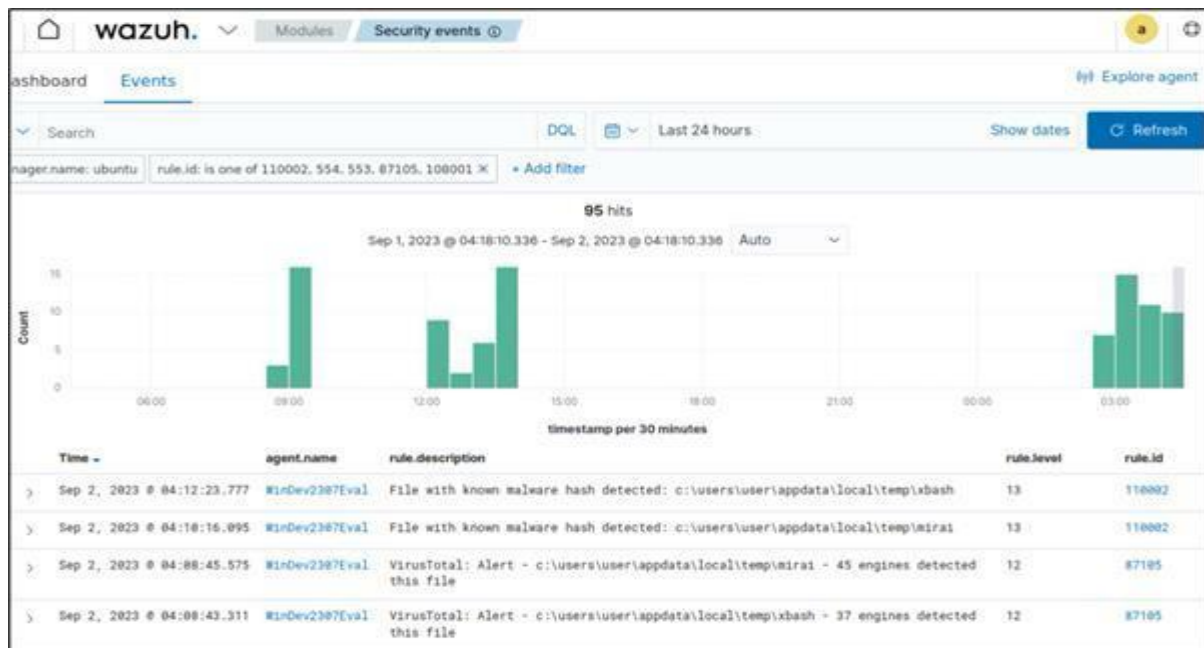


Fig.5 – Wazuh Dashboard

Real-time Wazuh logs indicate a malware detection event on the host system WinDev2307Eval, where suspicious files such as temp\ivbasha were flagged by VirusTotal and multiple antivirus engines as containing known malware signatures. These detections provide concrete evidence of file-based threats present on the system and are further supported by detailed correlations with specific rule IDs, detection timestamps, and alert severity levels, enabling security teams to trace the origin, assess the impact, and initiate timely remediation actions.



Fig.6 – Wazuh Log Analysis and Alert Summary

The SIEM dashboard presents a comprehensive real-time alert summary, currently displaying a total of 253 medium-severity alerts and 33 low-severity alerts, with no critical or high-severity threats detected at this time. This dashboard serves as a centralized security monitoring interface, integrating essential capabilities such as vulnerability detection, cloud security monitoring, threat hunting, and malware scanning, thereby enabling security analysts to effectively prioritize incidents, investigate potential threats, and maintain situational awareness across the entire infrastructure.

6. FUTURE ENHANCEMENT

To further strengthen the capabilities of this system, the following enhancements are proposed:

- **SIEM & SOAR Integration:** Integrate Wazuh with SIEM/SOAR platforms like Splunk or Cortex XSOAR for centralized threat visibility. Enable automated playbooks to streamline incident response and reduce analyst workload. This enhances detection, correlation, and real-time threat mitigation capabilities. Such integration supports faster, coordinated security operations across the enterprise.
- **Email & SMS Alerting:** Enhance Wazuh's alerting by adding multi-channel notifications via email, SMS, or messaging tools. Use platforms like Slack or Microsoft Teams to improve real-time collaboration during incidents. Custom notification rules ensure timely alerts based on event severity or type.
- **Threat Intelligence Feed Integration:** Incorporate external threat intelligence feeds such as AlienVault OTX or IBM X-Force Exchange into the Wazuh ecosystem. This will enable real-time enrichment of alerts with context about known malicious IPs, domains, and file hashes, helping analysts prioritize threats based on known indicators of compromise (IOCs). Such integration enhances proactive defense capabilities by correlating internal alerts with global threat data.
- **User Behavior Analytics (UBA):** Implement User Behavior Analytics to monitor and profile typical user activity across systems. By identifying deviations from established behavior baselines, the system can detect insider threats, compromised accounts, or privilege misuse that may not be flagged by traditional signature-based detection. Integrating UBA with Wazuh will add a behavior-driven detection layer, improving the system's ability to identify subtle or evolving threats.

7. CONCLUSION

In today's hyperconnected and increasingly cloud-driven environment, cybersecurity threats have become more dynamic, complex, and persistent. Among these, the use of malicious external IP addresses stands out as one of the most exploited vectors by cybercriminals. Attackers frequently use such IPs to perform reconnaissance, launch Distributed Denial of Service (DDoS) attacks, conduct phishing campaigns, carry out brute-force login attempts, or spread malware across corporate networks. This rapidly evolving threat landscape demands proactive, intelligent, and real-time defense mechanisms that go beyond traditional perimeter security models.

This internship sought to fill that gap by developing and deploying an external IP threat detection and response system in real-time based on the Wazuh open-source security monitoring platform, with the addition of external threat intelligence services such as VirusTotal. The system's primary objective was to continuously monitor incoming IP traffic, analyze behavioural patterns, and enrich threat data using external intelligence feeds in order to assess and score the risk level associated with each IP address.

A significant achievement of this work was the successful integration of VirusTotal threat intelligence into Wazuh's custom rule-based engine, allowing for automated decision-making based on IP reputation and behaviour. This integration eliminated the need for manual investigation of suspicious IPs and enabled the system to take responsive actions such as issuing alerts, logging the incident, and, where necessary, blocking malicious IPs via firewall rules. These actions were all visible and traceable through the Wazuh dashboard, providing transparency and accountability for every automated decision made.

The system not only demonstrates strong real-time threat detection capabilities but also supports scalability, flexibility, and cross-platform deployment, making it suitable for diverse environments—whether hosted in on-premises servers, cloud infrastructure, or hybrid networks. Its ability to automatically analyze and respond to threats based on evolving intelligence ensures that security teams can act faster, reduce false positives, and focus on higher-priority incidents.

In conclusion, this internship has demonstrated a cost-effective, automated, and intelligent threat detection system that enhances organizational cybersecurity resilience. It reinforces the idea that modern network defense is not just about monitoring traffic, but about correlating insights, adapting to real-time data, and responding intelligently and quickly to emerging threats.

REFERENCES

- [1] AbuseIPDB, "IP Address Reporting and Threat Intelligence," 2024.
- [2] Gartner, "Market Guide for Security Threat Intelligence Products and Services," Gartner Research, 2023.
- [3] IPinfo, "IP Geolocation and Intelligence API," 2024.
- [4] MITRE Corporation, "MITRE ATT&CK® Framework," 2024.
- [5] Open Web Application Security Project (OWASP), "OWASP Top 10 – Most Critical Web Application Security Risks," 2023.
- [6] VirusTotal, "VirusTotal API Documentation," Google, 2024.
- [7] Wazuh, "Wazuh Documentation," 2024.
- [8] S. Ramakrishnan and D. R. Chittibala, "Enhancing Cyber Resilience: Convergence of SIEM, SOAR, and AI in 2024," *Int. J. Comput. Eng.*, vol. 5, no. 2, pp. 36–44, 2024.
- [9] G. Areo, "Advancing Cyber Resilience through the Convergence of SIEM, SOAR, and AI Technologies," 2023.
- [10] S. Chatterjee, "Using SIEM and SOAR for Real-Time Cybersecurity Operations in Oil and Gas," *Int. J. Innov. Res. Comput. Technol.*, vol. 5, no. 1, 2020.
- [11] M. Hafiz and B. Soewito, "Information Security Systems Design Using SIEM, SOAR and Honeypot," *J. Pendidik. Tambusai*, vol. 6, no. 4, pp. 13856–13862, 2022.

- [12] C. S. Kummarapurugu, "Architectural Framework for Threat Intelligence Integration with SIEM and SOAR in Hybrid Cloud Security Environments," *Int. J. Innov. Res. Comput. Technol.*, 2024.
- [13] V. Hugo and M. Proust, "Integrating Firewalls with SIEM and SOAR Platforms for Automated Threat Response," *Int. J. Trend Sci. Res. Dev.*, vol. 6, no. 4, pp. 1–6, 2022.
- [14] M. Dacier, M. Almgren, and H. Debar, "Quality Evaluation of Cyber Threat Intelligence Feeds," in *Proc. ACNS*, 2020, pp. 264–284.
- [15] N. Rastogi, S. Dutta, M. J. Zaki, A. Gittens, and C. Aggarwal, "TINKER: A Framework for Open Source Cyberthreat Intelligence," arXiv preprint arXiv:2102.05571, 2021.
- [16] N. Rastogi, S. Dutta, M. J. Zaki, A. Gittens, and C. Aggarwal, "MALOnt: An Ontology for Malware Threat Intelligence," arXiv preprint arXiv:2006.11446, 2020.
- [17] V. Mavroeidis and A. Jøsang, "Data-Driven Threat Hunting Using Sysmon," arXiv preprint arXiv:2103.15194, 2021.
- [18] S. Mittal, A. Joshi, and T. Finin, "Cyber-All-Intel: An AI for Security related Threat Intelligence," arXiv preprint arXiv:1905.02895, 2019.
- [19] D. Schlette, "Threat Intelligence," in *Encyclopedia of Cryptography, Security and Privacy*, Springer, 2021.
- [20] Wikipedia, "Threat Intelligence," 2024. [Online]. Available: https://fr.wikipedia.org/wiki/Threat_Intelligence
- [21] Wikipedia, "MISP Threat Sharing," 2024. [Online]. Available: https://en.wikipedia.org/wiki/MISP_Threat_Sharing

- [22] Wikipedia, "Open Threat Exchange (OTX)," 2024. [Online]. Available: https://en.wikipedia.org/wiki/Open_Threat_Exchange
- [23] S. Ramakrishnan and D. R. Chittibala, "Enhancing Cyber Resilience through the Convergence of SIEM, SOAR, and AI in 2024," *Open J. Innov. J. Comput. Eng.*, vol. 5, no. 2, pp. 36–44, 2024.
- [24] Reddit, "Essential Log Sources for SIEM + SOAR," 2024. [Online]. Available: <https://www.reddit.com/r/cybersecurity/comments/17wfhvj>
- [25] Reddit, "SIEM & SOAR – What and how to automate?" 2024. [Online]. Available: <https://www.reddit.com/r/cybersecurity/comments/1eeuszn>
- [26] Reddit, "Good SIEM + SOAR solutions," 2024. [Online]. Available: <https://www.reddit.com/r/cybersecurity/comments/18m9pa7>
- [27] Reddit, "SIEM/SOAR or what?" 2023. [Online]. Available: <https://www.reddit.com/r/cybersecurity/comments/13x7183>
- [28] Reddit, "Threat Intelligence – How do you use it?" 2023. [Online]. Available: <https://www.reddit.com/r/cybersecurity/comments/ui6nqq>
- [29] GitHub, "awesome-threat-intelligence," 2024. [Online]. Available: <https://github.com/sk26mind/awesome-threat-intelligence>
- [30] Threat-Intelligence.eu, "Threat Intelligence Methodologies," 2024.